Fluid Mechanics:
Continuum
连续性假设与流体基本描述

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Contents

molecules $\Longrightarrow$ continuum field

$$\boxed{\rho = \rho(\boldsymbol{x}, t), \qquad \boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x}, t), \qquad p = p(\boldsymbol{x}, t)}$$

- microscopic:

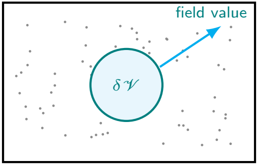
  molecules $\longrightarrow$ fluctuation

- local average:

  $$\Delta\mathscr{V} \longrightarrow \delta\mathscr{V}$$

- macroscopic:

  $\rho,\ \boldsymbol{u},\ p \longrightarrow$ smooth fields

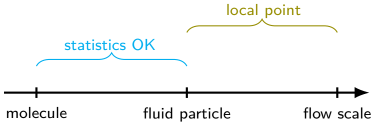


key: average over enough molecules, but keep the volume small compared with the flow scale.

fluid particle: not a molecule, not the whole flow

molecular scale $\ll$ fluid particle scale $\ll$ flow length scale

- many molecules: stable average
- small volume: local point
- moves with local variables



$$\rho(\boldsymbol{x},t)=\lim_{\Delta\mathscr{V}\to\delta\mathscr{V}}\frac{\Delta m}{\Delta\mathscr{V}},$$

$\delta\mathscr{V}\neq 0$ physically

$\mathrm{Kn} = \lambda/L$: criterion for continuum

$$\boxed{\mathrm{Kn} = \frac{\lambda}{L}}$$

λ : molecular mean free path

L : flow/geometric length scale

range	description	model idea
$\mathrm{Kn} < 10^{-3}$	continuum	Navier–Stokes usually OK
$10^{-3} \sim 10^{-1}$	slip/transition	boundary condition matters
$\mathrm{Kn} > 10^{-1}$	rarefied	continuum may fail

- continuum is not “molecules disappear”; it is a controlled averaging.

civil engineering scales: usually safe

- air at standard condition:
$$\lambda \sim 10^{-8}\ \mathrm{m}$$
- pipes, buildings, rivers:
$$L \gg \lambda$$
- micro gas flow: check Kn

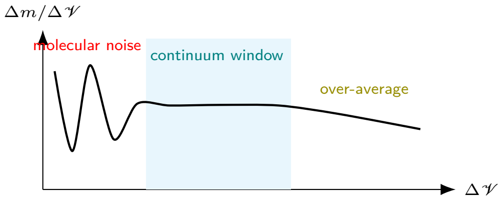
case	L	Kn
building wind	$10\ \mathrm{m}$	$\sim 10^{-8}$
ventilation duct	$0.1\ \mathrm{m}$	$\sim 10^{-6}$
micro-channel gas	$1\ \mu\mathrm{m}$	$\sim 10^{-2}$

$$\underbrace{\lambda \ll L}_{\text{scale separation}} \quad \Longrightarrow \quad \underbrace{\rho,\ \boldsymbol{u},\ p \text{ are fields}}_{\text{continuum model}}$$

density: local average

$$\rho(\boldsymbol{x}, t) = \lim_{\Delta\mathscr{V} \to \delta\mathscr{V}} \frac{\Delta m}{\Delta\mathscr{V}}$$

$\Delta\mathscr{V}$ too small $\Rightarrow$ fluctuation
$\Delta\mathscr{V}$ too large $\Rightarrow$ smearing
$\Delta\mathscr{V} \to \delta\mathscr{V}$ $\Rightarrow$ stable local value



velocity field: components and local value

$$\boldsymbol{u}(\boldsymbol{x},t) = u(\boldsymbol{x},t)\vec{e}_1 + v(\boldsymbol{x},t)\vec{e}_2 + w(\boldsymbol{x},t)\vec{e}_3$$

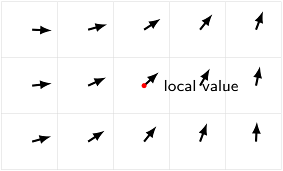
- vector value at every point:

$$\boldsymbol{x} = (x_1, x_2, x_3)$$

- acceleration is not only $\partial \boldsymbol{u}/\partial t$

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}\cdot\boldsymbol{\nabla})\boldsymbol{u}$$

$\boldsymbol{u}(\boldsymbol{x},t)$



Eulerian vs. Lagrangian

$$
\begin{aligned}
\text{Eulerian} &: \quad \boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x}, t) \qquad \text{fixed point in space} \\
\text{Lagrangian} &: \quad \boldsymbol{x} = \boldsymbol{x}(\boldsymbol{x}_0, t) \qquad \text{follow a fluid particle}
\end{aligned}
$$

Eulerian

- control equations
- boundary conditions
- CFD / PIV / probes

Lagrangian

- particle trajectory
- material derivative
- tracer particles

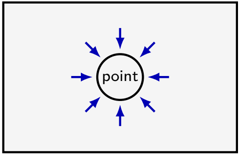
$$
\boxed{\frac{\mathrm{D}(\cdot)}{\mathrm{D}t} = \underbrace{\frac{\partial(\cdot)}{\partial t}}_{\text{local}} + \underbrace{\boldsymbol{u} \cdot \boldsymbol{\nabla}(\cdot)}_{\text{convective}}}
$$

pressure: normal force per unit area

$$p = \lim_{\Delta A \to 0} \frac{\Delta F_n}{\Delta A}$$

force $\perp$ surface

- hydrostatics: same pressure in all directions
- moving flow: pressure + viscous stresses
- wind load, pipe flow, outlet jet



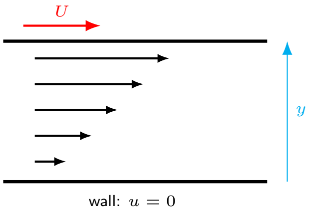
isotropic pressure at rest

viscosity: velocity gradient produces shear

$$\tau = \mu \frac{\mathrm{d}u}{\mathrm{d}y}$$

μ	:	dynamic viscosity
$\mathrm{d}u/\,\mathrm{d}y$	:	shear rate
τ	:	viscous shear stress

- no velocity gradient: no Newtonian shear
- viscosity creates boundary layers
- pipe friction and wake dissipation



material derivative

$$\frac{\mathrm{D}\phi}{\mathrm{D}t} = \underbrace{\frac{\partial \phi}{\partial t}}_{\text{unsteady at fixed point}} + \underbrace{\boldsymbol{u} \cdot \boldsymbol{\nabla} \phi}_{\text{change due to motion}}$$

for velocity: $$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = \frac{\partial \boldsymbol{u}}{\partial t} + u\frac{\partial \boldsymbol{u}}{\partial x} + v\frac{\partial \boldsymbol{u}}{\partial y} + w\frac{\partial \boldsymbol{u}}{\partial z}$$

- this is the bridge between Lagrangian thinking and Eulerian equations
- the nonlinear term appears because a particle moves through spatial gradients

incompressible: not “pressure cannot change”

$$\text{incompressible} \iff \frac{\mathrm{D}\rho}{\mathrm{D}t} = 0$$

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} = \frac{\partial \rho}{\partial t} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \rho, \qquad \text{mass conservation gives} \qquad \boxed{\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0}$$

- liquids: often incompressible
- low-speed gases: often incompressible when $\mathrm{Ma} < 0.3$
- pressure still varies; it enforces the velocity constraint

boundary condition: no slip

$$\boxed{\boldsymbol{u}_{\text{fluid at wall}} = \boldsymbol{u}_{\text{wall}}}$$

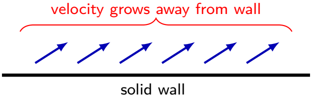
- stationary wall:

$$\boldsymbol{u} = 0$$

- moving wall:

$$\boldsymbol{u} = \boldsymbol{u}_{\mathrm{wall}}$$

- may fail for rarefied / micro-scale gas flows



when continuum fails

- high-altitude rarefied gas
- micro / nano channels
- strong non-equilibrium near shocks
- molecular-scale evaporation / condensation

$$\mathrm{Kn} = \frac{\lambda}{L} \quad \begin{cases} \ll 1, & \text{continuum} \\ \sim 1, & \text{kinetic effects} \end{cases}$$

failure $\neq$ bad equation failure $=$ bad local average

practice: estimate Kn

$$\lambda_{\text{air}} \approx 65 \text{ nm}, \qquad \mathrm{Kn} = \frac{\lambda}{L}$$

case	length L	Kn scale	judgment
building wind	10 m		
ventilation duct	0.1 m		
micro-channel gas	$1\ \mu$m		

question: why can the same air be continuum in buildings but questionable in micro-channels?

summary

- continuum assumption converts molecular systems into field problems
- fluid particle requires scale separation:

$$\text{molecule} \ll \text{fluid particle} \ll \text{flow scale}$$

- $\mathrm{Kn} = \lambda/L$ is the central criterion
- variables become fields:

$$\rho(\boldsymbol{x}, t), \quad \boldsymbol{u}(\boldsymbol{x}, t), \quad p(\boldsymbol{x}, t)$$

- next: control volume, mass conservation, continuity equation